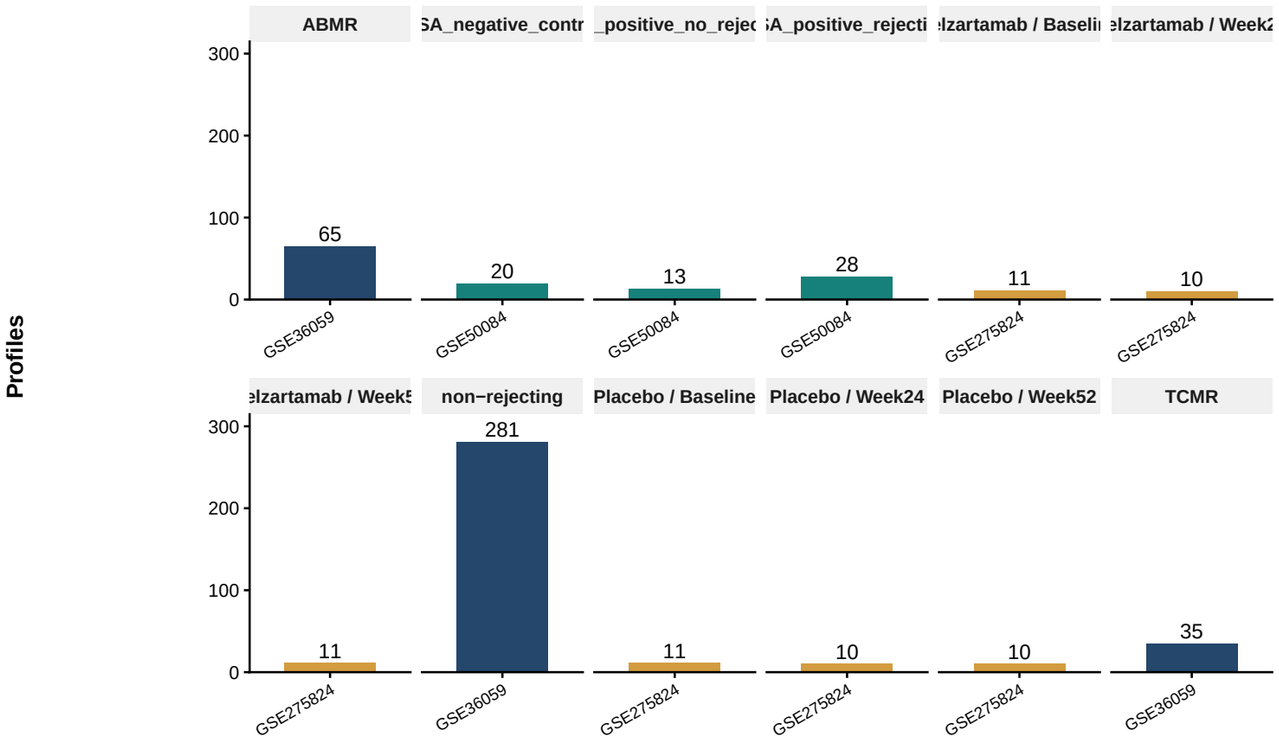
MOSAIC−Tx: reference−aware qualification of ABMR molecular signatures

Three public cohorts; within−cohort standardization; BH FDR within each declared contrast

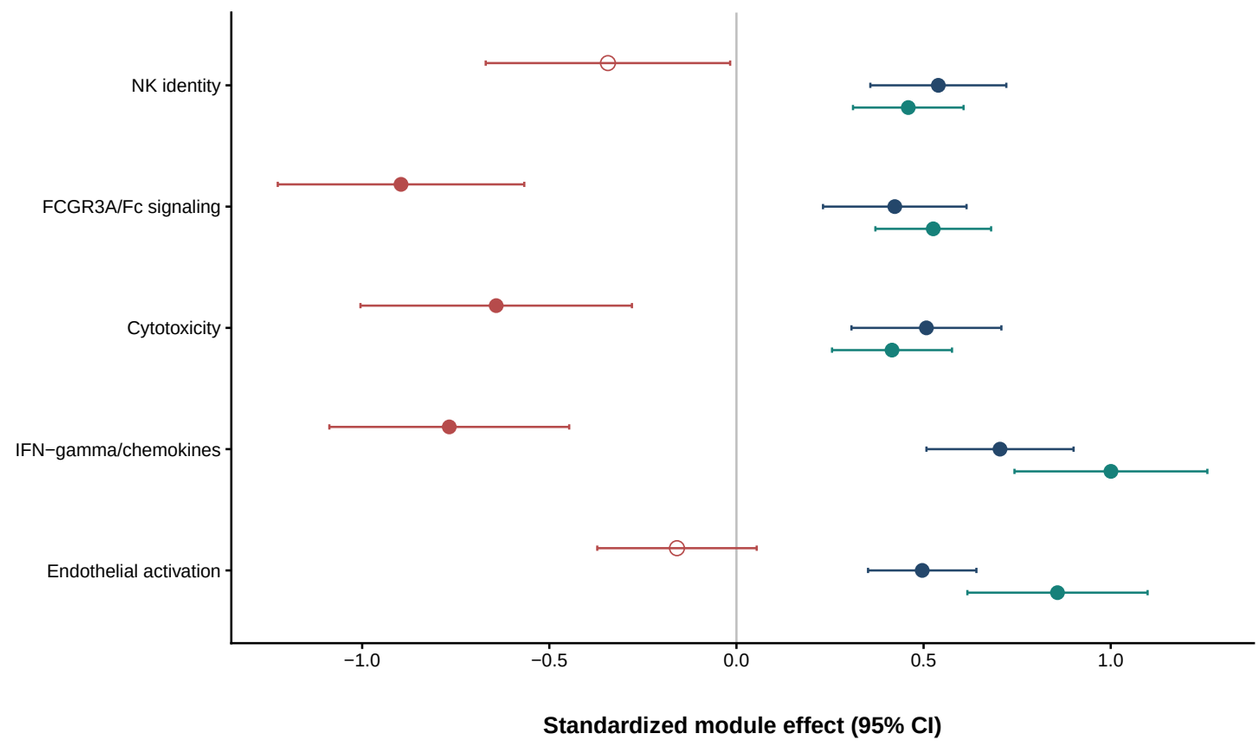
A Public−cohort architecture

Observed profiles entering each analysis



B Replication does not imply ABMR specificity

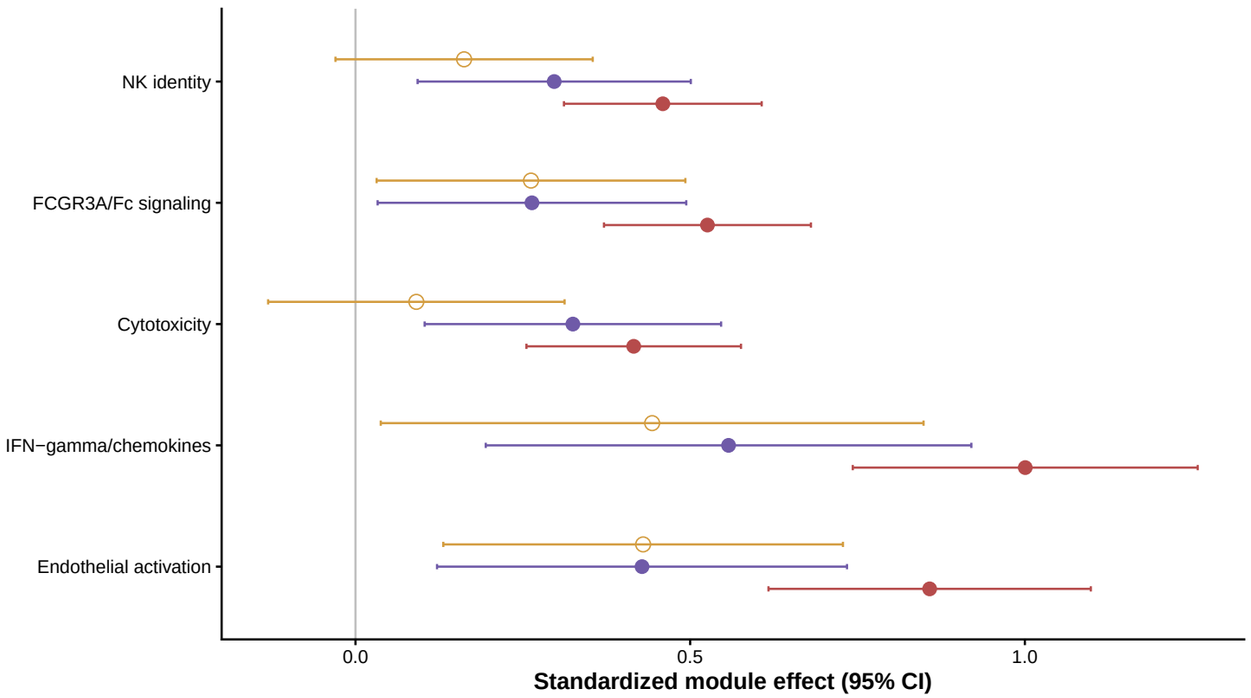
Positive effects favor ABMR; negative specificity effects favor TCMR



Evidence ● FDR < 0.05 ○ FDR >= 0.05 **Comparison** Discovery: DSA+ ABMR vs DSA− NR; Validation: ABMR vs NR

C DSA exposure is not equivalent to rejection

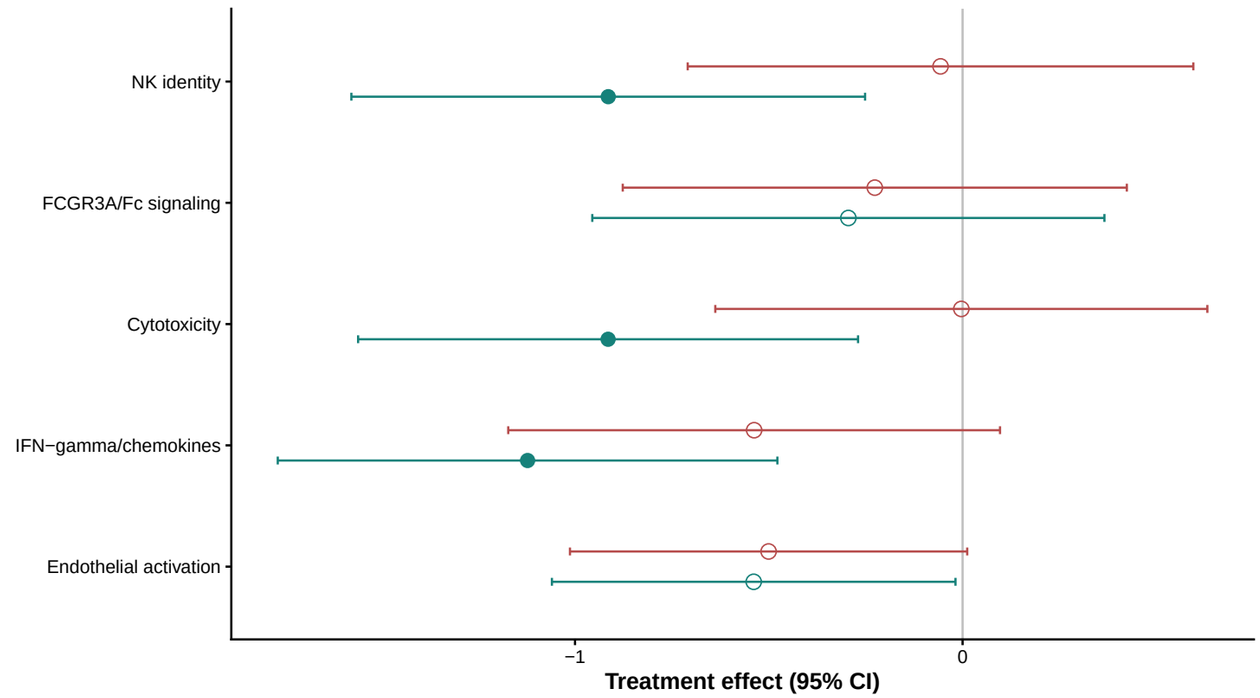
The DSA+ non−rejection contrast remains an explicit falsification test



Evidence ● FDR < 0.05 ○ FDR >= 0.05 **Comparison** DSA+ ABMR vs DSA− NR; DSA+ ABMR vs DSA+ NR; DSA+ NR vs DSA− NR

D Randomized treatment−associated module changes

Patient−blocked placebo−adjusted difference−in−differences



Evidence ● FDR < 0.05 ○ FDR >= 0.05 **Time point** Week 24; Week 52

NR, non−rejection. Effects are associative. Bulk expression cannot assign cell lineage or prove antibody engagement, target causality, or drug efficacy.